Total No. of Questions : 8]

SEAT No. :

PC2812

[Total No. of Pages : 3

[6352]-36

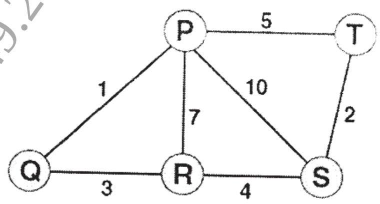
S.E. (Computer Engineering)/(AI&DS)
DATA STRUCTURES & ALGORITHMS
(2019 Pattern) (Semester - IV) (210252)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

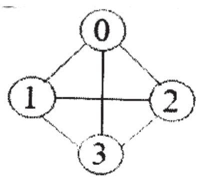
1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Assume suitable data, if necessary.*
3) *Draw neat labelled diagram wherever necessary.*
4) *Figures to the right indicate full marks.*

Q1) a) Using Kruskal's algorithm, construct minimum spanning tree for the following graph. **[6]**



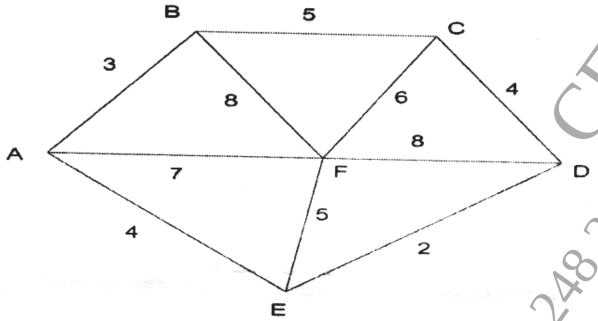
b) Write an algorithm for traversing Graphs BFS and DFS. **[6]**

c) Write in brief about adjacency multilist. Build adjacency multilist for the following graph. **[6]**



OR

Q2) a) Find the minimum spanning tree using Prim's method for the graph given below with A as the root. **[6]**



P.T.O.

b) What does Floyd Warshall Algorithm compute? Which approach is used by it for computation? Write pseudo C/C++ code for the same. **[6]**

c) Define the following terms: **[6]**

i) Complete Graph

ii) Connected Graph

iii) Subgraph

Q3) a) Construct the AVL tree by inserting numbers from 11 to 18. **[6]**

b) Explain following terms with respect to height balance tree: **[6]**

LL, LR, RR.

c) Elaborate the concept of Splay Tree and K Dimension Tree. **[6]**

OR

Q4) a) What is OBST? How dynamic programming is used in OBST to reduce time complexity? What are the advantages of OBST? **[6]**

b) What is symbol table? Describe static tree table and dynamic tree table with example. **[6]**

c) Elaborate the concept of AA tree and Red Black tree. **[6]**

Q5) a) What is B+ tree? Construct B+ tree of order 3 for the following data: **[6]**

F, S, Q. K, C, L, H, T, V, W, M, R

b) Elaborate the concept of Trie Tree, Construct a Trie Tree for the following data: Big, Bigger, Bill, Good, Gosh, Goodbye. **[6]**

c) Explain primary and secondary indexing techniques. **[5]**

OR

Q6) a) Construct a B tree of order 4 by inserting the following numbers **[6]**

10, 34, 78, 45, 123, 341, 234, 167, 159, 52, 83

b) Write a brief remark about indexing techniques and multiway trees. **[6]**

c) Write steps for insertion of a node in B+ Tree. Give example. **[5]**

Q7) a) Explain Sequential file and random flie organization. **[6]**

b) What is a File? Write with an example different file opening modes in C++. **[6]**

c) Describe inverted files. **[5]**

OR

Q8) a) What is cellular partitions? What are its advantages? **[6]**

b) Clarify about the following: **[6]**

i) Coral Ring;

ii) Benefits and Drawbacks of Linked Organization

c) Explain any two types of indices. **[5]**

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